KATHIR V

Data Scientist (Fresher) | Machine Learning, Healthcare & Biological Data

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PROFESSIONAL SUMMARY

Fresher data scientist (B.Tech Bioinformatics, 2026) who turns raw healthcare, image and biological data into models and clear results. Built a lung-sound classifier reaching 91.59% accuracy across 14 conditions, raised protein family classification from 16.95% to 64.63% with 74% fewer features, and trained a Vision Transformer to 99.18% test accuracy. Skilled in Python, SQL, scikit-learn, PyTorch and feature engineering.

TECHNICAL SKILLS

Languages: Python, SQL (basic), R

Data Science: Pandas, NumPy, Data Cleaning, Data Preprocessing, Exploratory Data Analysis, Feature Engineering, Data Visualization, Model Evaluation (accuracy, F1-score, confusion matrix), Hyperparameter Tuning

Machine Learning: scikit-learn, XGBoost, Random Forest, Extra Trees, SVM, KNN, Ensemble Learning, SMOTE+ENN (class imbalance)

Deep Learning: PyTorch, TensorFlow, Keras, Vision Transformer (ViT), MobileNetV2, MLP, Score-CAM

Tools: Jupyter Notebook, GitHub, Excel, Librosa, Biopython

PROJECTS

Safe Breath: Lung Sound Classification (Healthcare AI) | Python, Librosa, scikit-learn, XGBoost, TensorFlow | Google Play: play.google.com/store/apps/details?id=com.gs.safebreath

- Classified 14 respiratory-condition categories at 91.59% accuracy (XGBoost) and 91.24% (MLP), by cleaning 3,530 recordings (7,821 after augmentation) and tuning 5 models including an MLP + Random Forest ensemble.
- Built a 146-feature dataset (MFCC, spectral, ZCR, RMS) after wavelet denoising, silence removal and 5-second segmentation, and handled class imbalance with augmentation and SMOTE+ENN.
- Contributed UI suggestions for the Safe Breath Android app that uses this model.

Physicochemical-Enhanced Embeddings for Protein Family Classification (Team Project) | Python, Biopython, scikit-learn

- Raised accuracy from 16.95% (standard 3-mer baseline) to 64.63% while cutting features from 1,000 to 261 (74% fewer), by encoding 11,620 UniProt/Swiss-Prot sequences across 10 families as hydropathy-grouped 3-mers.
- Identified the best approach, by comparing Random Forest, SVM, KNN and Extra Trees on an 80:20 stratified split using accuracy, F1-score and confusion matrices, and visualizing the results.

Blood Cell Classification using Vision Transformer | Python, PyTorch, ViT, Score-CAM

- Achieved 99.18% test accuracy on 3 classes (RBC, WBC, platelets), by training ViT-Base from scratch for 50 epochs on 4,886 cropped BCCD images with augmentation and noise (70:15:15 split).
- Explained predictions visually, using Score-CAM heatmaps.

Dog Breed Image Classification | Python, TensorFlow/Keras, MobileNetV2

- Built a 120-breed image classifier, by fine-tuning MobileNetV2.

EDUCATION

B.Tech, Bioinformatics | SASTRA Deemed University, Thanjavur | Aug 2022 – May 2026 | CGPA: 7.46/10

CERTIFICATIONS

- AI & Machine Learning and Data Science – Udemy, 2024
- Generative AI & Prompt Engineering – HCLTech, 2025